

Progress report (Phase 2\_6th Semester) of the Final Year Project titled

# **Design and Implementation of an Automated Antenna Radiation Pattern Measurement System using Embedded Control and Image Processing**

*Submitted for partial*

*Fulfillment of the requirements for the degree of*

**Bachelor of Technology**

*in*

**Electronics and Communication Engineering**

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# CHAPTER:1

# Introduction

We introduce an automated system for precise antenna radiation pattern measurement, replacing error-prone manual methods. By integrating stepper-motor rotation, computer vision data acquisition, and CNC-based physical plotting, our project provides a low-cost, high-precision solution for RF characterization, ensuring repeatable results and seamless visualization for laboratories and educational institutions.

## I. Antenna Rotation Control system

The Antenna Rotation Control module ensures spatial accuracy using an ESP32 to drive a stepper motor in  $10^\circ$  increments. This automated synchronization captures precise radiation lobes and nulls while allowing fields to stabilize. By replacing manual positioning, the system guarantees a repeatable, scientific measurement cycle for high-resolution performance characterization.

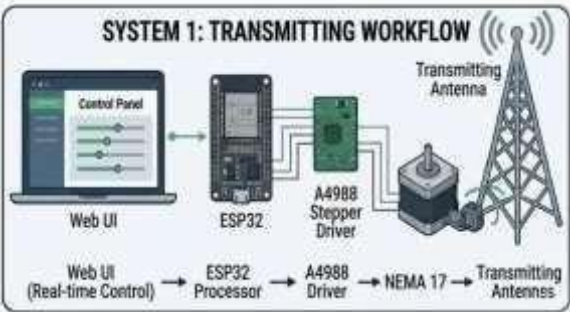


Fig 1.1:Antenna RotationControlSystem Workflow

## II. Image Processing Unit



Fig 1.2:Image ProcessingUnit Workflow

The Image Processing Unit uses an ESP32S-CAM for non-invasive data acquisition. By applying edge-based OCR to read measurement displays visually, it creates an "optical bridge." This ensures total galvanic isolation, eliminating electrical noise and "back-feeding" that often distort sensitive RF measurements, thereby maintaining the integrity of the electromagnetic environment.

## III. Physical Plotting System

The Physical Plotting System utilizes a 2-axis CNC plotter to translate signal data into tangible polar or Cartesian graphs. By automating the drafting process, it eliminates manual interpolation errors and human bias. The result is a professional, mathematically accurate hard-copy record of antenna gain and beamwidth with smooth, high-fidelity curves.



Fig 1.3:Physical PlottingSystem Workflow

## CHAPTER: 2

# **Review of previous work**

## **2.1 : Literature Survey:**

In the book Antenna Engineering Handbook, R. C. Johnson highlighted the limitations of manual radiation pattern measurement methods and emphasized the need for automated positioning systems to improve angular accuracy.

In the book Antenna Theory: Analysis and Design, Constantine A. Balanis discussed radiation pattern characterization techniques and measurement setups. He noted that traditional measurements require mechanical precision and suffer from human reading errors.

In the book Electromagnetic Waves and Antennas, S. C. Orfanidis introduced digital methods in RF testing and discussed how measurement automation can enhance repeatability and reduce observational inaccuracies in electromagnetic experiments.

In the book Handbook of Modern Sensors, J. Fraden explored non-contact sensing techniques, suggesting optical methods as reliable alternatives when direct electrical interfacing is not feasible.

Zhang et al. Proposed vision-based instrument reading systems using OCR to eliminate manual monitoring. Their research showed improved flexibility when image processing replaces wired data acquisition.

Y. Yalçın, S. Kurtulan developed motor-controlled positioning systems for antenna scanning that improved angular precision but lacked automated data extraction capability.

## **2.2 : Gap Identification:**

- ✓ Lack of Fully Integrated Automation: Existing systems automate either rotation OR measurement not both together.
- ✓ Dependency on Electrical Interface: Most systems require wired connection to meters, making them less flexible.
- ✓ Absence of Self-contained Embedded Operation: Many setups rely on PC-based processing.
- ✓ No Direct Physical Visualization: Graphs are digitally plotted but not physically drawn.
- ✓ Human Intervention Still Required: In positioning, reading, or plotting stage.

# CHAPTER: 3

## Theoretical Background

### 3.1 : Theory:

#### **Antenna Radiation Pattern Fundamentals**

A radiation pattern is a graphical representation of the radiation properties of an antenna as a function of space coordinates. Significance: It is critical in RF communication systems to determine the directionality, gain, and beamwidth of an antenna. Measurement Process: Traditionally, this involves rotating the antenna under test (AUT) and recording the received signal strength at specific angular intervals. Power Calculation: The system calculates relative power, often represented in decibels (dB), to accurately plot the signal intensity across different angles.

#### **Automation and Embedded Control**

To replace manual methods that are time-consuming and prone to human error, the project utilizes embedded systems for precise mechanical and logical control. Angular Precision: The system uses a NEMA 17 Stepper Motor coupled with an A4988 driver to achieve automated rotation in  $10^\circ$  steps. Centralized Processing: An ESP32 microcontroller serves as the "brain," managing the synchronization between the antenna's rotation, the data acquisition, and the plotting mechanism.

#### **Non-Contact Data Acquisition (Computer Vision)**

A key theoretical innovation in this project is the use of image processing to bridge the gap between legacy analog/digital meters and modern data logging. Optical Character Recognition (OCR): Instead of a direct electrical interface, an ESP32-CAM captures images of a meter display. Image Processing Pipeline: The system extracts numeric values from these images using OCR algorithms, allowing for "non-contact" data capture from any standard meter.

#### **Plotting Physical Visualization and**

Unlike standard digital systems that only produce virtual graphs, this project incorporates the theory of CNC (Computer Numerical Control) plotting. 2-Axis Translation: Using 28BYJ-48 stepper motors and ULN2003 drivers, the system moves a pen along X and Y axes. Direct Translation: The calculated relative power data is translated into physical coordinates to draw the radiation pattern directly on paper.

## CHAPTER: 4

### **4.1 : Circuit & Setup:**

### Circuit of Antenna Rotation Control system

The system is powered by an external 12V 2A adapter connected to the A4988 VMOT/GND pins, stabilized by a 100 $\mu$ F capacitor. Logic power (3.3V) and a common ground are provided by the ESP32.

Control is managed via GPIO 19 (STEP) for movement and GPIO 18 (DIR) for rotation direction, with the SLP and RST pins jumpered to maintain an active state. Finally, the A4988 drives the NEMA 17 motor through two coils: Coil 1 (Red/Deep Blue) and Coil 2 (Black/Sky Blue), ensuring precise, synchronized angular positioning for accurate RF signal capture.

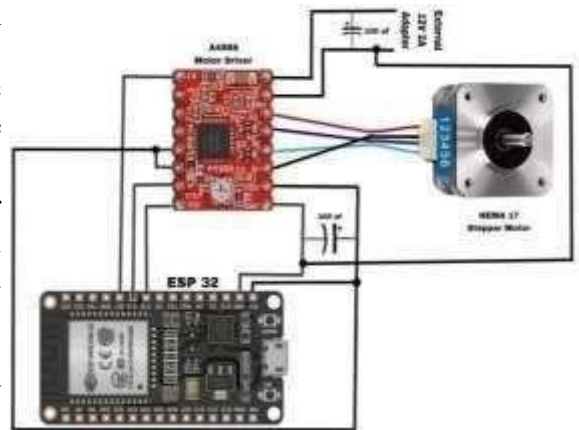


Fig4.1:Circuit of Antenna Rotation ControlSystem

## Setup of Image Processing Unit



- The USB cable provides a steady 5V supply from the laptop to the baseboard.
- The baseboard regulates this down to the 3.3V required by the ESP32 chip.

Fig 4.2: Setup of Image Processing Unit

## **4.2 : Hardware & Software Requirement:**

### **Hardware**

1. Central Processing Unit
  - o ESP 32 (512MB RAM) - Main controller
2. Mechanical Rotation System
  - o NEMA 17 Stepper Motor- Rotates antenna
  - o A4988 Motor Driver- Controls antenna rotation motor
3. Computer Vision and Image Acquisition
  - o ESP 32S CAM - Captures meter images
  - o ESP 32 CAM MB - Connector
4. Physical Plotting System
  - o 28BYJ-48 Stepper Motor(X-axis) – Horizontal movement
  - o ULN2003 Driver Board(X-axis) - X-axis motor control
  - o 28BYJ-48 Stepper Motor(Y-axis) - Vertical movement
  - o ULN2003 Driver Board (Y-axis) - Y-axis motor control
  - o Linear Rails(2 units) - 300mm aluminium guides
  - o Timing Belts- GT2 timing belts with pulleys
  - o SG90 Micro Servo - Pen lift mechanism
  - o Pen Holder- Spring-loaded mechanism
  - o Drawing Surface - A4 paper on rigid backing
  - o Frame Structure - Acrylic/aluminium frame
5. Power Supply System
  - o 12V 4A Power Adapter - USB-C connector for ESP 32

### **Software**

1. Arduino IDE
2. Visual Studio Code
3. C++, Python, HTML, CSS, JavaScript Programming Languages
4. NumPy, OpenCV, WiFi.h, esp\_camera.h, esp\_http\_server, Pytesseract Library Files
5. CP210X Windows Drivers

## 4.3 : Block Diagram:

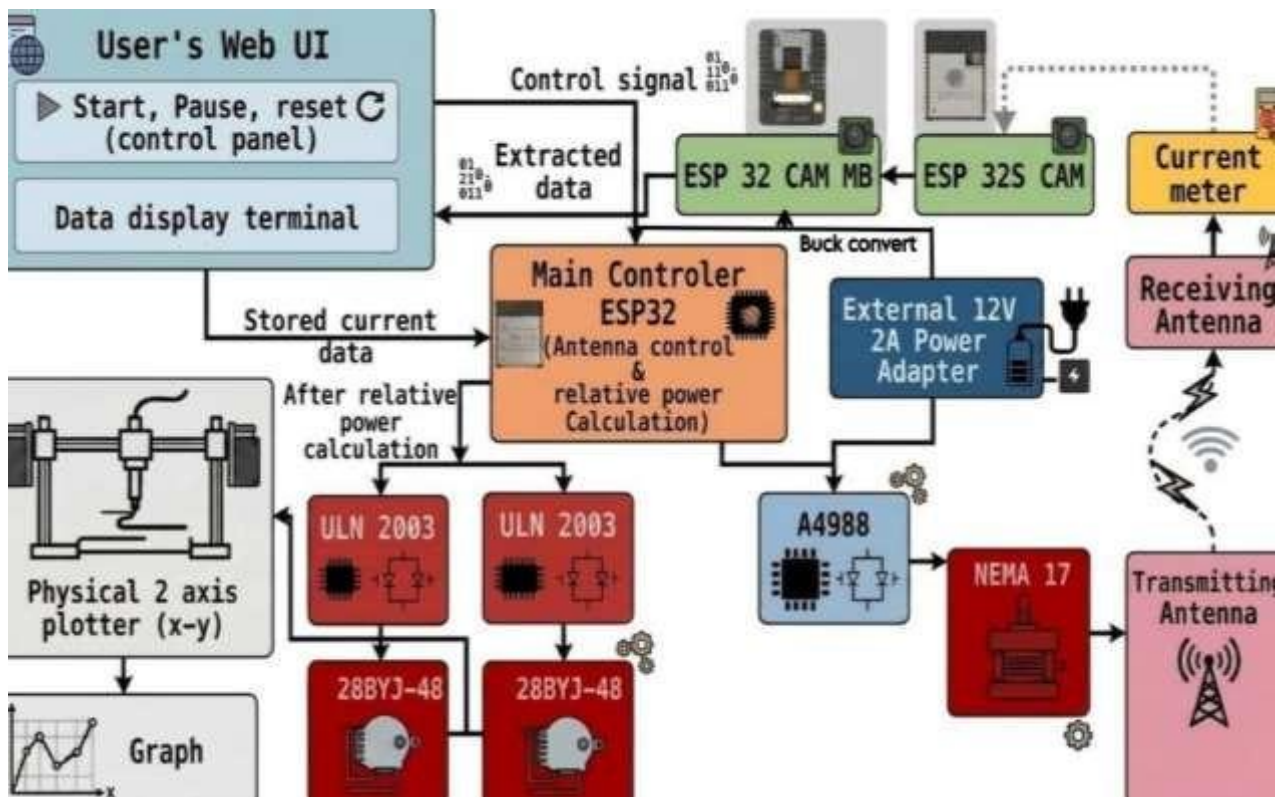


Fig 4.3: Block Diagram of Automated Antenna Radiation Pattern Measurement System

- ESP32 drives NEMA 17 stepper motor via A4988 driver
- Precise 10° angular increments (36 positions/full sweep)
- 30-second field stabilization pause at each step
- Web UI-based real-time control (Start / Pause / Reset)
- ESP32S-CAM captures meter display images via OCR
- Creates an "optical bridge" — complete galvanic isolation
- Eliminates electrical noise and back-feeding distortion
- Extracted values transmitted wirelessly to ESP32 controller
- 2-axis CNC plotter driven by 28BYJ-48 stepper motors
- ULN2003 driver boards control X and Y axes independently
- SG90 micro servo lifts/lowers pen for dot-free transitions
- Renders professional polar/Cartesian radiation graphs on A4 paper

## 4.4: Workflow:

- Setup & Initialize:

ESP32 boots, initializes Web UI server on local WiFi network.

- User Command (Web UI):

Operator selects Start/ Pause/ Reset from mobile or browser interface.

- Rotate Antenna (NEMA 17):

A4988 drives 10° step. System waits 30s for field stabilization.

- Capture & OCR (ESP32-CAM):

ESP32S-CAM photographs meter; OCR extracts current reading (e.g., 14.7871 mA).

- Calculate Relative Power:

ESP32 computes:  $P_{rel} = 20 \cdot \log_{10}(I/I_{max})$  in dB.

Stores data point.

- Repeat (36 positions):

Steps 3–5 repeat for each 10° increment across full

360° sweep.

- Physical CNC Plot:

28BYJ-48 motors plot the radiation graph on A4 paper via CNC plotter.

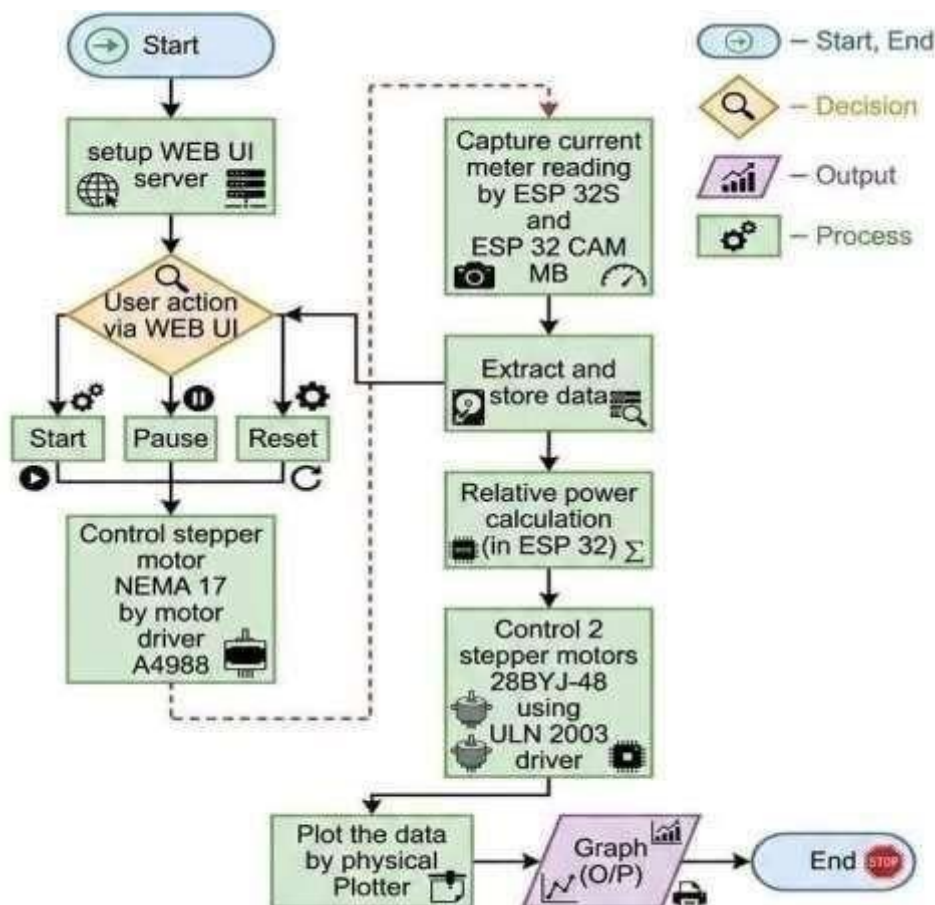


Fig 4.4: Workflow of Automated Antenna RadiationPatternMeasurement System

# CHAPTER: 5

## System Analysis

### 5.1 : Prototype Development:

#### Part 1: Embedded Rotation Control & Web UI

The ESP32-based control unit hosts a web server, providing a mobile "Antenna Command Center" for real-time operation. It drives a NEMA 17 motor via an A4988 driver in  $10^\circ$  increments. The system includes a 30-second stabilization pause at each step, ensuring high-precision, repeatable angular positioning and reliable data collection.

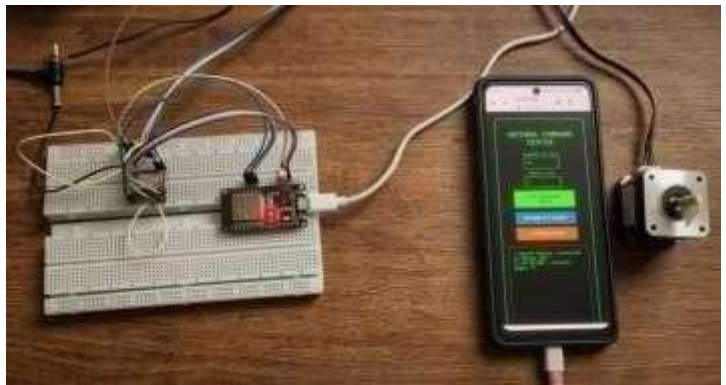


Fig 5.1: Embedded Rotation Control & Web UI

#### Part 2: Computer Vision & Data Acquisition

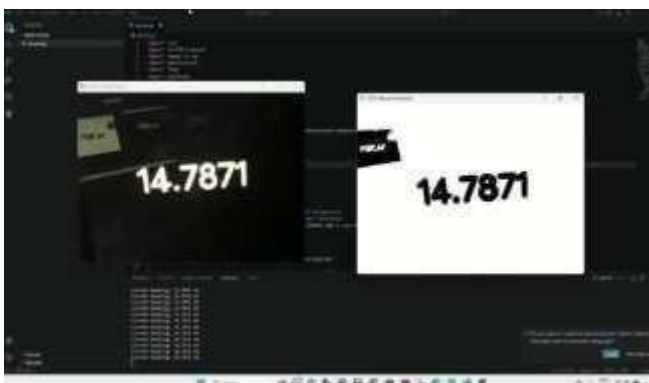


Fig 5.2: Computer Vision and Data Acquisition

The ESP32-S CAM unit facilitates non-invasive data acquisition by capturing high-resolution images of a meter's digital display. Using integrated OCR algorithms, the system extracts precise numerical values (e.g. "14.7871 mA" as seen in your prototype image) directly from the visual readout, providing complete galvanic isolation to prevent electrical noise from interfering with sensitive antenna measurement results.

## **5.2 : Future Scope:**

### **Native Polar Graph Plotting & Hard-Copy Output**

To align with standard industry conventions for antenna characterization, the system will transition from standard rectangular plots to true polar coordinate mapping. This update will allow for the direct generation of high-resolution, hard-copy radiation patterns. By representing gain and directivity in a circular format, users can more intuitively assess "front-to-back" ratios and beamwidth without manual data conversion.

### **AI-Enhanced OCR for Universal Meter Integration**

Reliability in varied environments is a primary goal. We intend to integrate a specialized deep-learning OCR model (based on a CNN or Transformer architecture) trained specifically on diverse LCD and LED meter displays. This will ensure robust data extraction even under challenging lighting conditions, high-glare environments, or skewed viewing angles, effectively making the system "meter-agnostic."

### **Multi-Frequency SDR Frontend Integration**

To move beyond fixed-frequency limitations, the next phase involves integrating a Software-Defined Radio (SDR) frontend. This will extend measurement capabilities across a vast RF spectrum. An SDR-based approach allows for real-time spectrum analysis, enabling the system to characterize wideband antennas and frequency-hopping IoT devices that traditional fixed-frequency meters might miss.

### **3D Spherical Radiation Pattern Mapping**

True antenna performance cannot be captured in a single plane. We plan to introduce a secondary elevation-plane scanning axis. This dual-axis movement, synchronized with the acquisition software, will enable full 3D spherical visualization. This allows developers to identify side lobes and nulls in a virtual 3D space, providing a comprehensive "heat map" of electromagnetic emission.

### **Strategic Market Integration**

The ultimate goal of these enhancements is to provide a high-value, automated alternative to six-figure professional RF instruments. By refining these features, we aim to serve three primary pillars of the RF community:

**IoT & Wireless Hardware Developers:** As the IoT market explodes, developers need precise, accessible tools to characterize small-form-factor antennas. Our 3D mapping and SDR support will provide the "pro-level" insights required for regulatory compliance and range optimization at a fraction of the cost.

**Automated Research & RF Labs:** By eliminating manual observation, we reduce the margin for human error and minimize "hand-effect" interference. These labs will benefit from the AI-enhanced OCR, allowing them to automate legacy equipment that lacks digital data outputs.

**Educational Institutions:** As a foundational tool for teaching electromagnetic theory, the transition to native polar plotting provides students with industry-standard visuals. This makes the system an ideal candidate for Educational Training Kit Manufacturers who provide hands-on RF curriculum to universities globally.

By converging these technical upgrades with a focused market strategy, the system is positioned to democratize high-fidelity RF measurement, turning complex electromagnetic theory into accessible, visual data..

# CHAPTER: 6

# Conclusion & References

## 6.1 : Conclusion:

Based on the current prototype development, the project has successfully achieved several critical milestones in automating antenna radiation pattern measurements:

**Antenna Rotation Control:** The system has successfully implemented precise antenna rotation using a NEMA 17 stepper motor.

**Web-Based Interface:** A functional web-based control interface has been developed and tested, allowing for real-time operation and motor control via a local server.

**Non-Contact Data Acquisition:** The integration of the ESP32-CAM and OCR-based monitoring system is complete, enabling the automated capture and extraction of numerical values from meter displays.

**System Integration:** Initial testing demonstrates successful communication between the hardware modules and software interfaces, proving the fundamental feasibility of the embedded automation system.

The technical integrity of this work is reinforced by high-quality documentation that ensures clarity and reproducibility. By detailing the problem statement, technology stack, and feasibility analysis alongside a thorough literature survey, the project establishes a rigorous academic foundation. The inclusion of detailed block diagrams for subsystems—such as motor control, image processing, and the plotting system—provides a clear roadmap of the design. Supported by technical justifications and references from IEEE papers and standard texts, this comprehensive reporting enhances the project's technical credibility and ensures that the system is both readable and reproducible for future research.

## 6.2 : References:

- 1 R. C. Johnson, Antenna Engineering Handbook, 3rd ed. New York, NY: McGraw-Hill, 1993.
- 2 C. A. Balanis, Antenna Theory: Analysis and Design, 3rd ed. Hoboken, NJ: Wiley-Interscience, 2005.
- 3 S. C. Orfanidis, Electromagnetic Waves and Antennas. New Brunswick, NJ: Rutgers University, 2002.
- 4 J. Fraden, Handbook of Modern Sensors: Physics, Designs, and Applications, 4th ed. New York, NY: Springer, 2010.
- 5 Zhang et al., 'Vision-based instrument reading using image processing and OCR techniques,' in Proc. IEEE Int. Conf. Systems, Man, Cybernetics, 2018.  
Link : <https://ieeexplore.ieee.org/document/8461184>
- 6 Y. Yalçın and S. Kurtulan, 'Motor-controlled antenna scanning system for radiation pattern measurement,' IEEE Antennas Propag. Mag., vol. 51, no. 3, pp. 147–155, Jun. 2009.  
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